

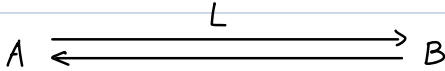
Speed of light

$$c = \frac{\Delta L}{\Delta t}$$

Galileo's experiment (1638),

Galileo di Vincenzo Bonaiuti de' Galilei

(1564 - 1642)



for $L \leq 1$ mile $\Delta t \sim 0$ (cannot tell) $\Rightarrow c \rightarrow \infty?$

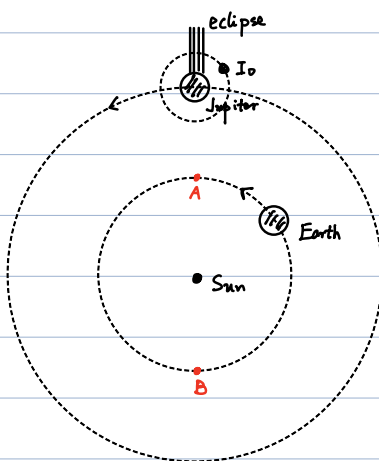
350 years: $\Delta t = \frac{3.2 \times 10^3 \text{ m}}{3 \times 10^8 \text{ m/s}} \approx 10^{-5} \text{ s}$

Rømer's observation

Ole Rømer (1644 - 1710)

Jupiter (木星) $\left\{ \begin{array}{l} \text{Zeus (Greek)} \\ \text{Jupiter (Roman)} \end{array} \right.$

Galilean moons $\left\{ \begin{array}{l} \text{Io} \\ \text{Europa} \\ \text{Ganymede (he)} \\ \text{Callisto} \end{array} \right. \rightarrow \text{lovers of Zeus}$



Orbital of Io: $T_0 \sim 42\text{h}$

Point A: Earth & Jupiter getting closer ($v_E > v_J$)

Period of observing eclipses: $T < T_0$

Point B: Earth & Jupiter getting further

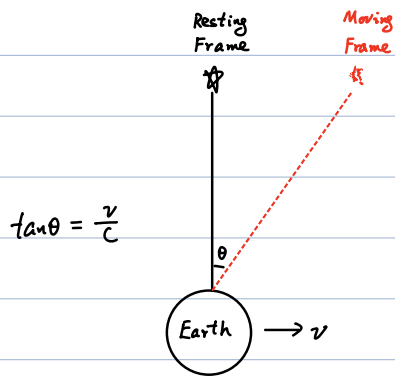
Period of observing eclipses: $T > T_0$

$\Rightarrow$ speed of light is finite! (Doppler)

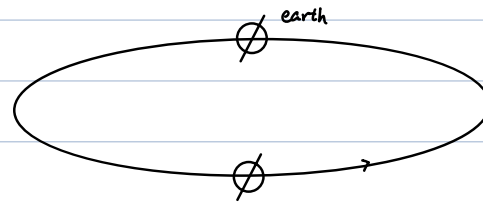
①

Aberration (光行差)

James Bradley (1693-1762)



↓ ↓ ↓ ↓ ↓ light from distant



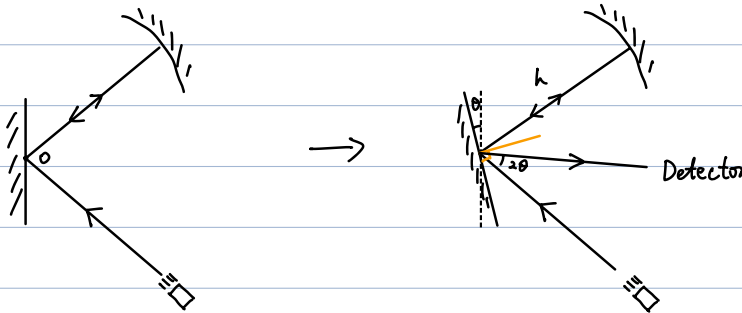
Seasonal change of θ .

$$\Rightarrow c \approx 3.01 \times 10^8 \text{ m/s} \quad (0.4\%) \quad 1729$$

Rotating mirror:

1862 $c \approx 2.98 \times 10^8 \text{ m/s} \quad (0.6\%)$

Léon Foucault (1819-1868)



$$\theta = \omega t$$

$t = 0$, light at O

$$t = \frac{2h}{c}, \quad \left\{ \begin{array}{l} \text{light back at } O \\ \theta = \omega \cdot \frac{2h}{c} \end{array} \right. \Rightarrow c = \frac{2\omega h}{\theta}$$

1926 $\left\{ \begin{array}{l} h \approx 35 \text{ km} \\ \text{improved apparatus} \end{array} \right.$

Albert Michelson (1852-1931)

$$\Rightarrow c \approx 2.99796 \times 10^8 \text{ m/s} \quad (0.001\%)$$

Violet

Red

Visible light: E&M wave with λ : 400 nm ~ 700 nm

$f = \frac{c}{\lambda} \Rightarrow f$: 7.5×10^{14} Hz ~ 4.3×10^{14} Hz

Q: A medium necessary for light?

"luminiferous aether", or "ether" 光以太, 以太
 substance that fills the space (including vacuum)

supporters

- 4th-century BC, Aristotle.
- 17th century, Boyle, Huygens, Newton, ...
- 19th century, A lot of questions...

• Longitudinal wave $v = \sqrt{\frac{B}{\rho}}$ B: bulk modulus

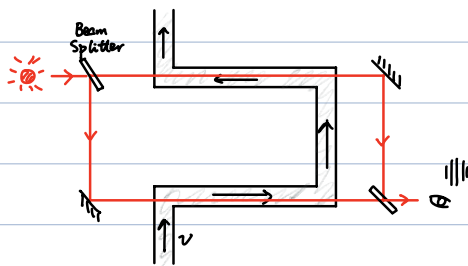
✓ Transverse wave $v = \sqrt{\frac{G}{\rho}}$ G: shear modulus

c is large $\Rightarrow$ G is large (ether is hard) $\Rightarrow$ hard to move around...

Fizeau Experiment (1851)

Armand Fizeau (1819 - 1896)

Speed of light in moving water:



Naively, flowing water "drags" ether

$$\Rightarrow \begin{cases} v_+ = \frac{c}{n} + v \\ v_- = \frac{c}{n} - v \end{cases} \begin{array}{l} \text{ether moves with water} \\ \end{array}$$

n : index of refraction
 $n \approx 1.333$ for water

Measured: $v_+ = \frac{c}{n} + v \left(1 - \frac{1}{n^2}\right)$

$$n = 1.333 \Rightarrow \left(1 - \frac{1}{n^2}\right) \approx 0.44$$

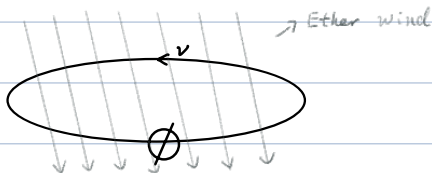
much lower drag speed...

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What about air?

$$n \approx 1 \Rightarrow v \approx \frac{c}{n}$$

air flows with earth's rotation $\Rightarrow$ No drag of ether by earth? $\Rightarrow$ ether wind?

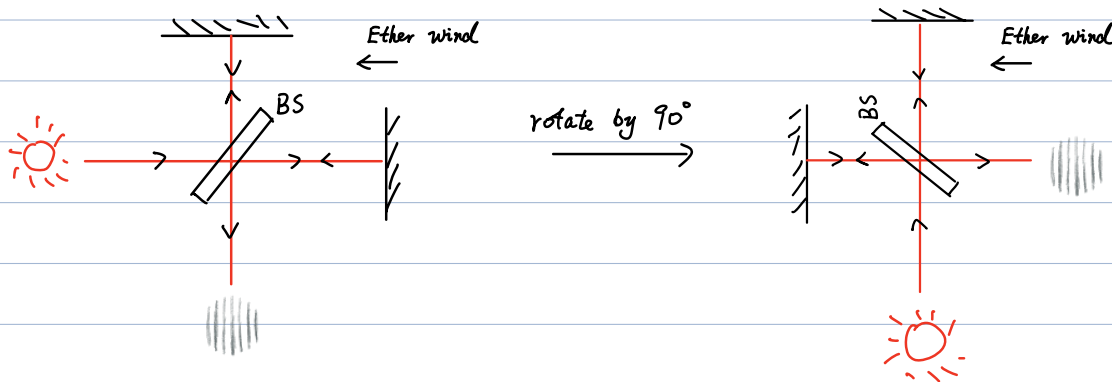


Relative speed between earth & ether: $v \sim 3 \times 10^4$ m/s (ether wind)
(if ether static)

Michelson-Morley Experiment (1887)

"Most famous 'failed' experiment"

Nobel Prize 1907



Expect: drift of interference by 0.4 fringes

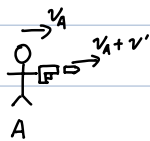
Experiment: ≤ 0.01 fringes

$\Rightarrow$ Ether is stationary relative to earth (complete drag) $\Rightarrow$ no ether wind?

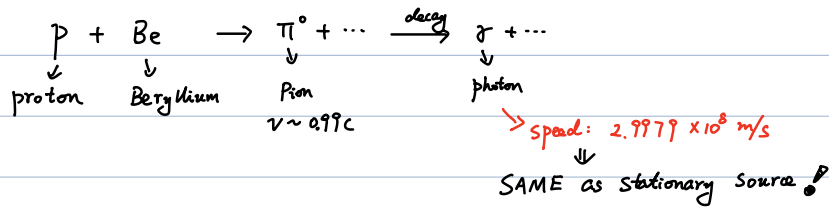
Q: No drag (Fizeau) vs. Complete drag (Michelson-Morley)?

$\Rightarrow$ Ether is self-contradicting!

Moving source:



Moving light source (Alväger et al. 1964)

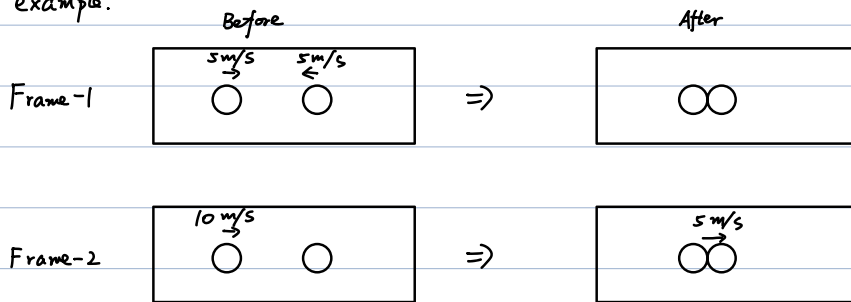


Important assumptions in relativity

1) Physical laws in ALL inertial frames are the same.

(cannot tell if an inertial frame is static or not)

example.



Which frame is static? Cannot tell...

• If ether is present,

$\Rightarrow$ a special inertial frame (ether static)

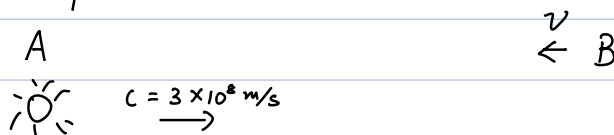
$\Rightarrow$ ugly ...

2) speed of light is the same in all inertial frames.

(A brave guess by Einstein)

Albert Einstein (1879-1955)

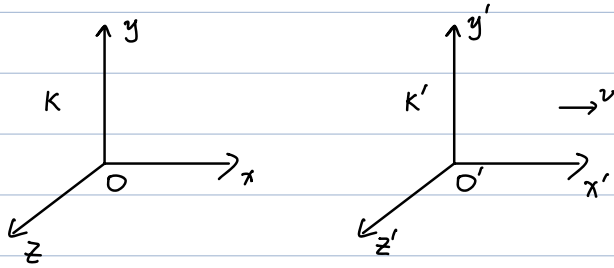
Example.



In frame static wrt B, c still $3 \times 10^8 \text{ m/s}$?

⑤

Lorentz transformation



At $t = t' = 0$, light source at $O = O'$

in K frame: spherical wavefront

in K' frame: still spherical?

$$\begin{cases} x^2 + y^2 + z^2 - c^2 t^2 = 0 & (\text{in } K) \\ x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0 & (\text{in } K') \end{cases}$$

$\Rightarrow$

$$x^2 + y^2 + z^2 - c^2 t^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$$

No relative motion along y & z: $y' = y, z' = z$

$$\Rightarrow x^2 - c^2 t^2 = x'^2 - c^2 t'^2$$

Try a linear transformation:

$$\begin{cases} x' = x \cosh \varphi - ct \sinh \varphi \\ ct' = -x \sinh \varphi + ct \cosh \varphi \end{cases} \quad (\cosh^2 \varphi - \sinh^2 \varphi = 1)$$

$$\begin{aligned} \text{verify: } x'^2 - (ct')^2 &= x^2 \cosh^2 \varphi + c^2 t^2 \sinh^2 \varphi - 2xct \sinh \varphi \cosh \varphi \\ &\quad - (x^2 \sinh^2 \varphi + c^2 t^2 \cosh^2 \varphi - 2xct \sinh \varphi \cosh \varphi) \\ &= x^2 - c^2 t^2 \end{aligned}$$

let $\beta \equiv \tanh \varphi$

$$\Rightarrow \begin{cases} \sinh \varphi = \frac{\beta}{\sqrt{1-\beta^2}} \\ \cosh \varphi = \frac{1}{\sqrt{1-\beta^2}} \end{cases}$$

$$\Rightarrow \begin{cases} x' = \frac{x - \beta ct}{\sqrt{1 - \beta^2}} \\ t' = \frac{t - \beta \frac{x}{c}}{\sqrt{1 - \beta^2}} \end{cases}$$

$$\begin{cases} 0 \text{ in } K \text{ frame: } x=0 \\ 0 \text{ in } K' \text{ frame: } x' = -vt' \end{cases}$$

$$\Rightarrow \begin{cases} -vt' = \frac{-\beta ct}{\sqrt{1 - \beta^2}} \\ t' = \frac{t}{\sqrt{1 - \beta^2}} \end{cases}$$

$$\Rightarrow \beta = \frac{v}{c}$$

Lorentz transformation:

$$\begin{cases} x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \\ y' = y \\ z' = z \\ ct' = \frac{ct - \frac{v}{c}x}{\sqrt{1 - v^2/c^2}} \end{cases} \quad \begin{cases} x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}} \\ y = y' \\ z = z' \\ ct = \frac{ct' + \frac{v}{c}x'}{\sqrt{1 - v^2/c^2}} \end{cases}$$